

Manual for updating the atrial fibrillation rhythm control treatment sequence model

Version 1.0

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1 Introduction

1.1 Purpose of the manual

This document serves as support for adaptations to other countries or updates of the treatment sequence model for rhythm control in patients with atrial fibrillation. **Please read the technical report or scientific publication of the model before starting your update of the model.**

1.2 Collaborators

The treatment sequence model for rhythm control in patients with atrial fibrillation was developed in 2024-2025 in a collaboration with Zorginstituut Nederland, Cochrane Netherlands (for the systematic literature review and meta-analysis) and the 'Nederlandse Vereniging voor Cardiologie (NVVC)'. Find an overview of the collaborators in the table below.

Organisation	Name of involved staff
Zorginstituut Nederland	Matthijs Versteegh Simone Huygens Marijke Delsing
Cochrane Nederland (Julius)	Demy Idema Anneke Damen Kevin Jenniskens Lotty Hooft
NVVC	Martin Hemels Dominik Linz Michiel Rienstra Sing-Chien Yap

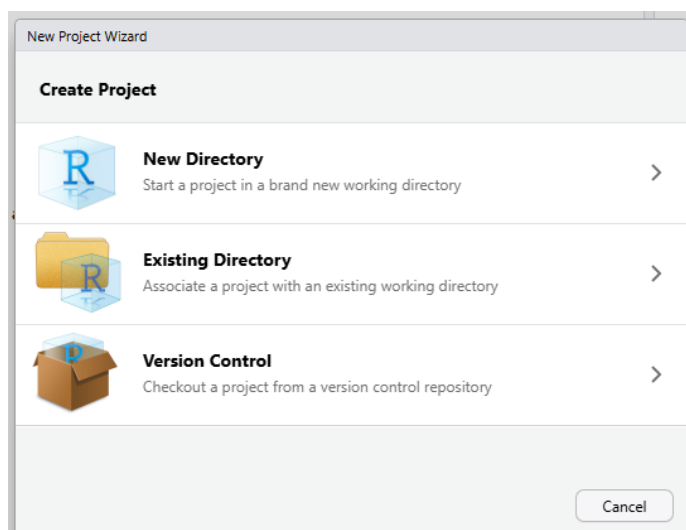
1.3 Prerequisites and required software/tools

Adapting or updating the model code requires access to and experience with R and R Studio and Git/Github.

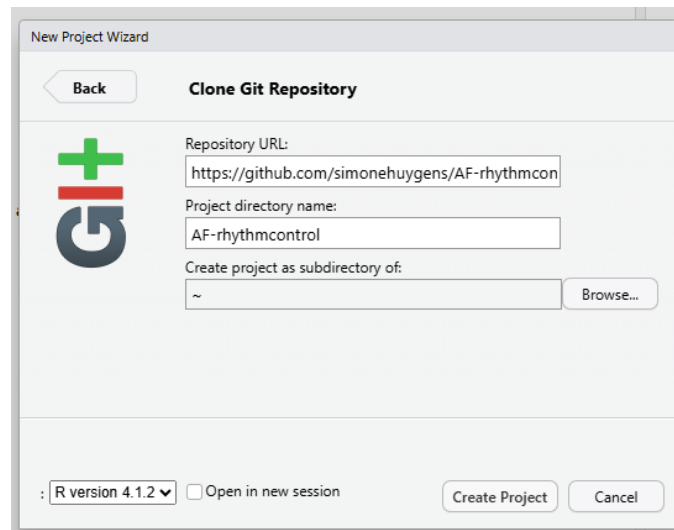
1.4 Cloning the model into R from Git

Please follow the following steps to load the model in to the R Studio environment.

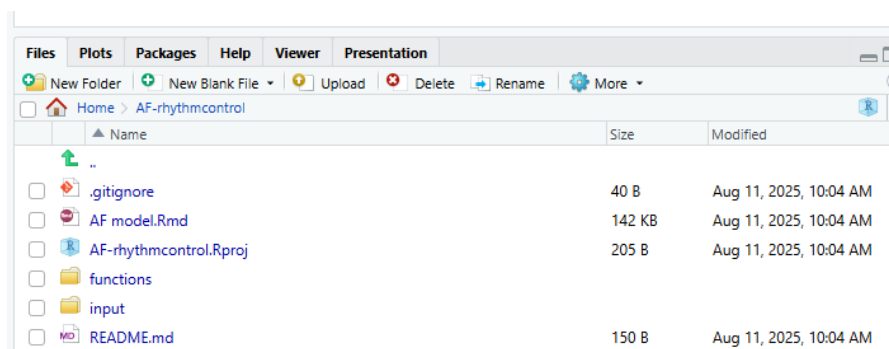
1. Open RStudio
For Zorginstituut: Open the Posit workbench via the Microsoft Edge browser and select '+ New Session'.
2. Click on 'File' in the top left and then on 'New project' to open a new project.
3. You will see the menu below, select 'Version Control' and then 'Git'.



4. Fill in the Git repository link of the AF model at 'Repository URL':
<https://github.com/simonehuygens/AF-rhythmcontrol.git>
 It is a public git repository, so no password is needed.
5. The 'Project directory name' will then be filled in automatically.



6. Click on 'Create Project' and the project and associated files will be loaded.
7. In the bottom right (usually, but depending on your own RStudio layout) under the 'Files' tab you will now see the files for the AF model. Open the 'AF model.Rmd' file.

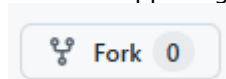


The 'input' folder contains all necessary input files. The 'functions' folder contains the 'functions.R' script that is needed to run the model and is automatically loaded when running section 1.3 of the 'AF model.Rmd'.

1.5 Log code changes or 'fork' the model

Please make a log of the code changes that have been made. Ideally, start a new repository on Github with the old code as source (using the fork option on GitHub) and the updates as new commits so that version control is possible all the way back to the original model. This is an alternative to cloning (described in 1.4).

To do this you can click the Fork button on the upper right of the [repository's GitHub page](#).



GitHub will create a complete copy of the model under your own account with your own username/repository-name version. You can make changes, commit and push to your fork freely.

2 Adapt or update input parameters

2.1 Settings

Section 1.2 describes the main settings of the model. Most settings should not be adapted. Only the following settings can be changed or adapted as needed:

- The minimum age (min_age) should not be lower than 18, because the model is developed for adults.
- The maximum age (max_age) should be equal to the maximum age in the background mortality table.
- The number of patients (n_i) should be at least 50,000 for stable model results, but for practice run this can be reduced to increase the speed of the model simulations.
- For country adaptations, the willingness to pay threshold (wtp), discount rate for costs (d_rc) and discount rate for effects (d_re) can be adapted.

```
# Model settings
min_age <- 18 # min age: only include adults
max_age <- 108 # max age based on max in Human Mortality Database
...
n_i <- 50000 # number of simulated individuals
...
wtp <- 20000 # willingness to pay threshold
d_rc <- 0.030 # discount rate costs
d_re <- 0.015 # discount rate effects
```

2.2 Patient population

The characteristics of the patient population are set in section 2.

Users can adapt the mean and standard deviation of age, the proportion of females and the proportion of paroxysmal versus persistent AF. Based on these inputs, the model will generate a patient population with varying ages and sex.

When changing the age, the calculation of the weighted average of age of survived and deceased patients can be removed, as this was only included for the specific source (de Mol et al. 2022) that was used for age in the original model.

```
#### PATIENT CHARACTERISTICS ####
n_survived <- 30182 # de Mol et al. 2022 (NHR 2013-2020) Table 1
n_deceased <- 15 # de Mol et al. 2022 (NHR 2013-2020) Table 1
n_total <- n_survived+n_deceased
age <- ((61.4*n_survived)+(65.9*n_deceased))/n_total
sd_age <- ((9.8*n_survived)+(7.1*n_deceased))/n_total
n_female <- 9816+7
n_male <- n_total-n_female
prop_female <- n_female/(n_female+n_male)
n_paroxysmal <- 19527+11 # proportion paroxysmal AF
n_persistent <- 7300+3+542+0 # proportion persistent AF
prop_parox <- n_paroxysmal/(n_paroxysmal+n_persistent)

# Generate patient population (will be varied in PSA)
set.seed(2)
v_age <- round(rtruncnorm(n_i, min_age, max_age, age, sd_age))
v_Sex <- rbinom(n_i, 1, prop_female)
```

2.3 Clinical input parameters

The model consists of the following clinical input parameters:

Input parameter	Source
Relative risks of AF recurrence AAD vs. CA	Meta-analysis Cochrane Nederland
Probability of recurrence of AF after CA after 1 year	Meta-analysis Cochrane Nederland and health insurance claims data from Vektis
Long-term recurrence of AF symptoms after CA	Health insurance claims data from Vektis
Background mortality	Human Mortality Database
Excess mortality	Vinter et al. 2020
Discontinuation of AAD's due to adverse events	Meta-analysis Cochrane Nederland
Adverse events CA	NHR ablation analysis tool (Behandelgroep: Katheterablatie AF, year: 2023)

This chapter explains how these input parameters can be adapted.

2.3.1 Relative risks and short term AF recurrence

The relative risks (RR_nai, RR_AAD_exp_parox, RR_AAD_exp_pers, RR_CA_exp*) and probabilities of recurrence after 1 year (prop_AF_CA_nai, prop_AF_CA_AAD_exp_parox, prop_AF_CA_AAD_exp_pers, prop_AF_CA_CA_exp_1*, prop_AF_CA_CA_exp_2*) are mainly based on the meta-analysis. The results of the meta-analysis are loaded in section 2 of the .Rmd. This loads the data that is needed to calculate the relative risks of AAD versus CA and the probabilities of recurrence of AF 1-year after the CA (see code below).

```
#### Probabilities of treatment success ####
# Load results of meta-analysis Cochrane Nederland
load(here::here("input", "meta-analysis-efficacy.RData"))
load(here::here("input", "meta-analyse disc_AAD.RData"))
```

To update these RData objects, the original scripts of the meta-analysis need to be updated. However, this script and underlying data extraction file is not publicly available. Users can manually adapt the parameters if they are provided in other sources (for example a publication of a new meta-analysis) or contact Cochrane Netherlands to update the meta-analysis with the same methods as the original meta-analysis. Please be aware that for the OWSA you ideally need the lower and upper limit of the 95% confidence intervals and for the PSA you need the standard error of the input parameters.

*Currently these input parameters were not available in the meta-analysis, therefore they were derived from health insurance claims data.

2.3.2 Long-term AF recurrence

These input parameters are based on the health insurance claims data that is loaded from the input folder using the following code.

```
load(here::here("input", "TTE_AF_Vektis.RData")) # survival model objects
(version 17-04-2025)
```

If an update of the data is available, this RData object needs to be updated. This RData object consists of three flexsurvreg objects called best_fit_ablation 1 to 3. This contains the details of the best fitting parametric distribution of the Kaplan-Meier curve of the first, second and third or higher CA estimated on the health insurance claims data.

These can be replaced with flexsurvreg objects that are estimated on updated data or other data sources.

For Zorginstituut:

If an update of the model is required, contact Marijke Delsing to update the data file

generated on 17 April 2025. Then follow the following steps to update the flexsurvreg objects:

1. Save the 'alle' tab in a separate .csv file.
2. Save the .csv file in the 'input' folder of the R project using the upload button in the right bottom panel of R studio.



3. Open the R script 'Update Vektis time to AF recurrence analysis.R' in the 'functions' folder of the R project.
4. Change the file name of the .csv file in the R script to the new file name.
5. Run the R script 'Update Vektis time to AF recurrence analysis.R' with the updated .csv file. This will automatically save the best_fit_ablation objects in the 'input' folder under the name that is specified at the bottom of the R script.
6. Change the file name of the .RData file to the new file name.

```
if(scenario == "basecase"){
  load(here::here("input", "TTE_AF_Vektis.RData")) # survival model
  objects (version 17-04-2025)
```

2.3.3 Mortality

To update or change the source for background mortality in the general population create a .csv file with a column for age ("Age") and columns for mortality rates for females ("Female") and males ("Male").

Save the .csv file in the 'input' folder of the R project using the upload button in the right bottom panel of R studio.



Change the reference to the .csv file in the code in section 2 of the .Rmd (see below). Note that you may have to change the separator in read.csv. Additionally, the mortality rate of the highest age in the mortality table should be set to 1. The code to do this is already provided (see the bottom of the code below), but you need to make sure 108 is the maximum age and if not adapt it to the maximum age in your mortality table.

Probability of death

Background mortality in the general population weighted by the sex distribution of the patient population

We used 2019 to avoid bias from COVID-19

```
df_mort <- read.csv(here::here("input", "sterfte_NL_2019.csv"),
  sep = ";", header = TRUE)
```

```
df_mort <- df_mort[, -c(1,5)] # remove year and total column
```

```
df_mort <- df_mort %>% # convert table from wide to long with a variable for sex
```

```
  pivot_longer(
    cols = c(Female, Male),
    names_to = "Sex",
    values_to = "r_mort"
  ) %>%
  mutate(
    Sex = ifelse(Sex == "Female", 1, 0),
    r_mort = as.numeric(r_mort)
  )
```

```
df_mort$r_mort_c1 <- df_mort$r_mort*c1 # convert annual mortality rate to x-weekly mortality rate
```

```
df_mort$p_mort_c1 <- 1-exp(-df_mort$r_mort_c1) # convert x-weekly mortality rate to probability
```

```
df_mort[df_mort$Age == 108, "p_mort_c1"] <- 1 # set mortality rate to 1 at max age to make sure everyone dies in the model
```

The hazard ratio for excess mortality can be changed here:

```
HR_EM <- 2.01 # 95% CI: 1.71 to 2.36 (Vinter et al. 2020)
```

2.3.4 Adverse events

For AADs adverse events that result in discontinuation of AADs are included in the model. The probability to discontinue is derived from the meta-analysis and can be updated in section 2 (see below).

Probability of severe adverse events

Discontinuation of AAD's due to adverse events

```
p_disc_AAD_nai <- 0.121 # Source: meta-analysis Cochrane Netherlands
```

```
p_disc_AAD_exp <- 0.225 # Source: meta-analysis Cochrane Netherlands
```

For CA three complications: cardiac tamponade, phrenicus paralysis and vascular complications are included in the model. The number of events and total number of CAs can be updated in section 2. If these are not available the probabilities can directly be imported in the p_CT_CA, p_PP_CA and p_VC_CA variables.

Adverse events CA (Source: event rates from NHR ablation analysis tool: Behandelgroep: Katheterablatie AF, Jaar: 2023)

```
n_CT_CA <- 26 # cardiac tamponade
```

```
n_tot_CT_CA <- 6150
```

```
p_CT_CA <- n_CT_CA/n_tot_CT_CA
```

```
n_PP_CA <- 25 # phrenicus paralysis
```

```
n_tot_PP_CA <- 6211
```

```
p_PP_CA <- n_PP_CA/n_tot_PP_CA
```

```
n_VC_CA <- 40 # vascular complications
```

```
n_tot_VC_CA <- 6149
```

```
p_VC_CA <- n_VC_CA/n_tot_VC_CA
```

2.4 Costs

Healthcare costs are loaded in df_c_states, see below:

Health state costs and costs of interventions (Source: Vektis)

```
df_c_states <- load(here::here("input", "c_Vektis.RData"))
```

This loads the following input parameters:

Data	
🔵 c_AAD_params	List of 17
🔵 c_CA_DBC_params	List of 17
Values	
c_AAD	51.289397993311
c_CA_DBC	11105.5982207358
c_d_after_CA1	643.063444816054
c_d_after_CA2	93.5856722408027
c_d_before_CA	739.286528428094
c_d_SAF	830.249444816053
c_sd_d_after_CA1	6163.50101959616
c_sd_d_after_CA2	4500.5095504739
c_sd_d_before_CA	4392.83348669121
c_sd_d_SAF	4476.6291761755
df_c_states	chr [1:13] "c_d_SA
n_c_Vektis	1495L

These input parameters and associated standard deviations can be changed manually if other data sources are available. Please be aware that variables that start with c_d are cost

differences compared to no AF symptoms.

For Zorginstituut:

If an update of the model is required, contact a business analyst (e.g. Marijke Delsing) to update the data file. Then follow the following steps to update input parameters:

1. Save the .xlsx file in a .csv file.
2. Save the .csv file in the 'input' folder of the R project.
3. Open the R script 'Update Vektis cost analysis.R' in the 'functions' folder of the R project.
4. Run this script with the updated .csv file and save the results in an .RData file in the 'input' folder (the code to do this is already provided at the end of the script).

Societal cost calculations can be changed in the code if new data becomes available.

2.5 Correct costs for inflation

The costs are corrected for inflation with the `adjust_inflation` function in the 'functions.R' script. If the costs need be updated with a new correction for inflation, the `target_year` and inflation rate for future years can be changed in the 'functions.R' script. The `target_year` should be changed to the current year. The `infl_rates` for the following year can be added under 2023. NB. The inflation rate for 2023 is actually the rate that is reported at 2024 at 'Jaarmutatie CPI afgeleid (%)' in [CBS](#) (but it is the inflation rate from 2023 to 2024 and is therefore reported with "2023" here).

```
#### Adjustment for inflation using consumer price index ####
# Cost year = 2024, except for drug costs (2025)
# Correct for inflation
adjust_inflation <- function(amount, source_year, target_year = 2024) {
  # Define yearly inflation rates (as decimals)
  infl_rates <- c(
    "2017" = 0.014,
    "2018" = 0.016,
    "2019" = 0.012,
    "2020" = 0.025,
    "2021" = 0.118,
    "2022" = 0.030,
    "2023" = 0.024
  )

  # Get years to include in calculation
  years <- as.character(source_year:(target_year-1))

  # Calculate inflation factor and apply
  factor <- prod(1 + infl_rates[years])
  return(amount * factor)
}
```

2.6 Utilities

The utility input parameters consists of three elements: the disutility of having AF symptoms (`du_SAF`), the disutility for complications after CA (`du_CT_mo`, `du_PP_mo`, `du_VC_mo`), and the utility without AF symptoms which is assumed to be equal to the general population utility (`mod_splines`).

The disutility of having AF symptoms or complications after CA can be replaced with another value if there is another source available or preferred.

The general population mortality should be adapted to the local setting if possible. In the

original version of the model it is included in a regression formula with age and sex as covariates. In the Effs function (in section 03.6) this model is used with the predict function to calculate the utility associated with the patients current age and sex. The mod_splines object can be replaced with another R object of a regression.

UTILITIES INPUTS

Disutilities AF and adverse events

Load utilities based on the EQ-5D-5L data in the AVATAR-AF trial using the Dutch tariff

```
load(here::here("input", "glm_AVATARAF_NL.RData"))
```

```
du_SAF <- abs(unnname((glm_AVATARAF_NL$coefficients[2]))) # Re-analysis of AVATAR-AF with EQ-5D-5L NL tariff based on Moss et al.
```

Assumption similar to Akerborg et al. (2012) and Reynolds et al. (2014)

```
du_CT_mo <- 0.1 # Assumption disutility of 0.1 for 1 month, corrected to disutility for the cycle length in Effs
```

```
du_PP_mo <- 0.1 # Assumption disutility of 0.1 for 1 month, corrected to disutility for the cycle length in Effs
```

```
du_VC_mo <- 0.1 # Assumption disutility of 0.1 for 1 month, corrected to disutility for the cycle length in Effs
```

General population utilities

Regression model general population by age and sex (based on Versteegh et al. 2016)

```
load(here::here("input", "splines_HRQoL.RData"))
```

```
gen_pop_utility <- "fit"
```

```
mod_splines_coef <- mod_splines$coefficients
```

3 Functions

The model uses several functions for the microsimulation:

- Create_df_X: creates a dataframe with patient characteristics (section 3.1).
- PrepareProbs: calculates all required clinical input parameters for the model from the clinical input parameters defined in section 2 (section 3.2).
- Probs: creates the transition probability matrix for each patient (section 3.3) and is based on the DARTH modelling framework.[refs]
- PrepareCosts: calculates all required cost input parameters for the model from the cost input parameters defined in section 2 and adjusts the inputs for inflation if necessary (section 3.4).
- Costs: calculates the costs per cycle for each patient (section 3.5) and is based on the DARTH modelling framework.[refs]
- Effs: calculates the QALYs per cycle for each patient (section 3.6) and is based on the DARTH modelling framework.[refs]
- MicroSim: combines all functions to run the microsimulation (section 3.7) and is based on the DARTH modelling framework.[refs]

For more information on the individual functions we refer to the annotations in the .Rmd and/or the DARTH modelling framework literature.[refs]

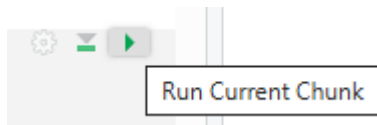
4 Run analyses

After the model is adapted, the model can be used to run analyses.

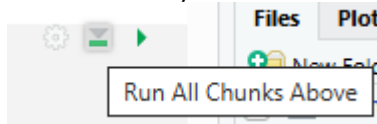
For basic users, we advise to use the 'AF model.Rmd' to run the model.

When using the 'AF model.Rmd', we advise to either:

- run the chunks one by one manually by clicking on 'Run Current Chunk'



- or to go to section 04.1 and click on 'Run All Chunks Above' to load the complete model and the input parameters. Then run 04.1 or any of the other chunks below to run an analysis with the model.



- It is not recommended to run all chunks from top to bottom of the .Rmd consecutively because this includes sensitivity analyses that have a long runtime and there is a higher likelihood of mistakes.

In 4.1 the analysis can be run for one sequence. The treatments TRT1 to TRT6 can be adapted by the user and should either be "CA" or "AAD".

```
outcomes <- MicroSim(1_params, n_i, df_X, TRT1 = "CA", TRT2 = "AAD", TRT3 = "AAD", TRT4 = "AAD", TRT5 = "AAD", TRT6 = "CA", seed = 1)
```

There are several analyses available in the .Rmd:

- Deterministic analysis of 1 sequence (section 4.1)
- Deterministic analyses of all sequences (section 4.2), including health state and line traces and summary tables.
- One-way sensitivity analyses (section 5), including tornado diagrams of costs, effects and the net health benefit.
- Probabilistic sensitivity analyses (section 6), including creating cost-effectiveness planes, cost-effectiveness acceptability curves, and value of information analysis.
- Model stability testing (section 7) which can estimate the number of individuals (n_i) that should be simulated to have stable deterministic results.

In addition to the .Rmd, the 'functions' folder contains scripts to run a base-case analysis, scenario analysis, one-way sensitivity analysis or probabilistic analysis and to test the model stability. These scripts can be useful when you want to start a 'Workbench job' in the Posit Workbench which enables you to run several analyses at the same time while still being able to use R for other analyses. Please make sure to adapt the 'model functions.R' script (or any of the other scripts) with the changes you made in the 'AF model.Rmd' in case you want to use any of these scripts to run analysis.